Data Integration for ML-CNPM$_{2.5}$: A Public Sample Dataset Based on Machine Learning Models and Remote Sensing Technology Applied for Estimating Ground-Level PM$_{2.5}$ in China

Yulong Fan, Lin Sun, and Xirong Liu

***Abstract*— Ambient fine particulate matter (PM$_{2.5}$) has significant adverse effects on human health, thereby urgent hunger for accurate monitoring of ground-level PM$_{2.5}$, especially its space distribution. Since satellites can observe the Earth on a large spatial scale, remote sensing technology can be applied to estimate PM$_{2.5}$ concentrations at the national level. Based on it and machine learning (ML) methods, numerous studies mapped high-accuracy, wholesale and continuous PM$_{2.5}$. However, different models and data in these studies made their results incomparable, and more samples were needed to be provided. Here, a large-column and long-term sample dataset (ML-CNPM$_{2.5}$) applied for ML-based models was constructed with 5 076 608 data records and 24 features from 2014 to 2023 in China. Multiple approaches were used to guarantee the quantity and quality of the sample dataset. Due to its comprehensiveness and objectivity, the ML-CNPM$_{2.5}$ can be used to train and validate different models, thereby further improving the accuracy of PM$_{2.5}$ estimating. Using the ML-CNPM$_{2.5}$, eight basic ML-based models were also constructed as the baseline for judging other derivative models. These models can estimate daily full-coverage PM$_{2.5}$ and most performed well, with ten-fold cross-validation (10 CV) RMSE of 16.94–11.21 μg/m^3 and R^2 of 0.71–0.89, which is consistent with previous studies and can effectively capture spatial trends of PM$_{2.5}$ in a period suffered from high pollution. Overall, our ML-CNPM$_{2.5}$ can be applied to effectively construct, validate, and compare various ML-based models for PM$_{2.5}$ estimation, helping to develop new algorithms with higher accuracy and robustness.**

***Index Terms*— Aerosol optical depth (AOD), machine learning (ML), particulate matter (PM$_{2.5}$), sample dataset, satellite remote sensing.**

I. Introduction

EXPOSURE to high-level fine particulate matter (PM$_{2.5}$, particulate matter's aerodynamic diameter less than 2.5 μm) can harm human health, leading to increased morbidity and mortality related to respiratory diseases, especially for the older and children [1], [2], [3]. In 2015, about 6.5 (95% confidence interval (CI) 5.7–7.3) million population dead are attributable to air pollution risk factors according to the Global Burden of Disease study (GBD), especially 4.2 (95% CI 3.7–3.8) million due to ambient particulate, which is about 1–4 times higher than other air pollution (i.e., household air and ambient ozone pollution) from the report named The Lancet Commission on pollution and health [4]. In nations with rapid economic growth, such as China, severe air pollution, particularly from PM$_{2.5}$, is a significant issue. It was estimated that around 851,660 deaths were attributed to ambient PM$_{2.5}$ pollution in China in 2017 [5] alone. Therefore, obtaining PM$_{2.5}$ information promptly and precisely is of significance for controlling and reducing PM$_{2.5}$ concentration at the national level, especially for developing countries like China.

Optical satellite remote sensing has currently turned into a major technology for obtaining ground-level PM$_{2.5}$ maps due to its long-term and big-scale observation-earth ability [6], [7], [8], [9]. Satellite observation can extend ground-based PM$_{2.5}$ measurements and even reconstruct PM$_{2.5}$ records in the periods when the ground stations were not established, prompting research on the impact of ambient PM$_{2.5}$ on human health and climate [10], [11], [12]. Methodologically, machine learning (ML) models, e.g., random forests (RF), extreme gradient boosting (XGBoost), support vector machine (SVM), light gradient boosting machine (LGB), k-nearest neighbor (KNN), and deep belief network (DBN), have been widely used to estimate ground-level PM$_{2.5}$ and generate spatiotemporal continuous products spanning country to globe due to their powerful information-extract ability [7], [8], [10], [12], [13]. For example, Yan et al. [14] proposed a novel Spatial–Temporal Interpretable Deep Learning Model (SIDLM) aiming to improve the interpretability and accuracy of ground-level PM$_{2.5}$ estimating during 2016–2019 in China. Chen et al. [15] used an improved ML-based RF model to estimate PM$_{2.5}$ concentration in Shenzhen, China,

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from 2016 to 2018. Wei et al. [8] presented using the Space–Time Extra-Trees (STET) model based on extremely randomized trees (ERT) to estimate ground-level $PM_{2.5}$ over mainland China and released a historical $PM_{2.5}$-record dataset spinning from 2000 to 2018 based on their proposed model, filling the gap of no national monitoring $PM_{2.5}$ data before 2013. Furthermore, the data used in the previous studies was highly heterogeneous in time, space, and sources; thereby, the results from different studies are incomparable, especially regarding their proposed models' accuracy and prediction power.

Therefore, comparisons between different ML-based models should be conducted to determine the optimal model demonstrating the best performance on ground-level $PM_{2.5}$ estimating based on remote sensing technology. Considering this issue, Li et al. [16] validated and compared the accuracy of four models, i.e., orthogonal regression (OR), regression tree (RT), RF, and SVM, in estimating ground-level $PM_{2.5}$ over the Beijing–Tianjin–Hebei (BTH) region of China in 2015. Furthermore, Ma et al. [17] used six ML-based models, i.e., multiple linear regression (MLR), KNN, SVM, RT, RF, and backpropagation neuron network (BPNN), to estimate ground-level $PM_{2.5}$ under different scenarios to find the best model according to the coefficient of determination (R^2). Their results indicate that the various ML-based models performed distinctly in estimating $PM_{2.5}$ concentration under different conditions (i.e., regions or periods). Due to the different regions or date ranges when most previous studies were carried out, however, the final results between them are also incomparable. Therefore, a comprehensive and objective sample dataset covering the same time and space is urgently needed as the standard to validate the $PM_{2.5}$ results of different ML-based models, ensuring comparability between models. In contrast to other ML-based retrieval tasks using satellite remote sensing, e.g., Cloud Detection task (CD) and Land Classification task (LC), ground-level $PM_{2.5}$ retrieval lacks a harmonized and accessible public dataset (PD) as a reference to evaluate models, especially those new ML-based models. Both CD and LC possess ample PDs, e.g., LandSat8-Biome [18], CloudSEN12 [19], UC Merced Land Use [20], and AID [21] are available to validate the performance of the models on their respective tasks. The above difference between $PM_{2.5}$ retrieval and the two remote sensing tasks can be attributed to the difference in the final target regarding classification targets for CD and LC and regression results for $PM_{2.5}$ retrieval. It is easier to construct a comprehensive and ample dataset for classification compared to regression tasks like $PM_{2.5}$ because the reference targets of classification can be obtained through definitions and judgment from different experts, whereas it is challenging for $PM_{2.5}$ retrieval to collect reference targets under certain circumstances. For instance, ground-based $PM_{2.5}$ measurements were unavailable in most cities of China before 2013, when a national network of $PM_{2.5}$ monitoring stations had just begun to be established, meaning insufficient data for validating and evolving $PM_{2.5}$ retrieval models early, especially ML-based models requiring massive samples [6], [7], [8]. Fortunately, up to now, a total of over a decade (2013–2024) of ground-based $PM_{2.5}$ measurements from various stations across different cities in China can be collected. This valuable data, thus, can be used as a reference target to more comprehensively and appropriately validate, compare, and evolve ML-based modes than ever before.

Here, we construct the first comprehensive PD, namely ML-CNP$M_{2.5}$, through matching ground-based $PM_{2.5}$ measurements with simulation data, meteorological data, geographic data, and economic data from 2014 to 2023 over China, ensuring training, validation, and comparison of ML-based models for ground-based $PM_{2.5}$ retrieval. Additionally, emulating the validation of ML-based models in the large language modeling domain and computer vision field, we also validated and compared multiple basic ML models, including tree-based models (i.e., RF, ERT, XGBoost, LGB, and AdaBoost) and neuron-based models (i.e., DBN, FCNN, and ResNet), a total of eight models, as the baseline to evaluate other derivative models proposed in subsequent studies. Section II describes the data methods used to construct ML-CNP$M_{2.5}$ and ML-based models compared and validated based on our proposed PD. Then, the results, including objective verification of ML-CNP$M_{2.5}$ and ML-based models, are demonstrated in Section III. Next, we discuss the potential applications of the ML-CNP$M_{2.5}$ and the pros and cons of the basic models validated combined with previous research in Section IV. Finally, Section V summarizes the main content and purpose of this work.

II. Materials and Methodologies

A. Data Sources

1) Ground-Based $PM_{2.5}$ Measurements: Ground-based $PM_{2.5}$ measurements obtained from China National Environmental Monitoring Center (CNEMC) during 2014–2023 in mainland China [73.66–135.05°E, 3.86–53.55°N] are treated as reference targets to construct our PD, ensuring training and validating ML models fully. These ground-based $PM_{2.5}$ are measured through tapered element oscillating microbalance (TEOM) (or beta gauge instruments located) in ground stations, with accuracy about 1.5 $\mu g/m^3$ (0.1 $\mu g/m^3$) [22] and has been widely applied for validation of $PM_{2.5}$ retrieval [7], [8], [23], indicating that they can be used as a data source of reference target for our sample dataset due to their high quality. In this study, the data used were calibrated and processed [24] before being applied to make up our PD, ensuring the precision of the sample data. Our PD not only includes daily average $PM_{2.5}$ of station measurements but also momentary average $PM_{2.5}$ covering Terra satellite overpass moments (9:00–12:00 A.M. local time), supporting ML-based models' generation and validation for different applications [24]. The average $PM_{2.5}$ values greater than 500 $\mu g/m^3$ are excluded in this study for further accuracy. The detailed information about ground-based $PM_{2.5}$ measurements is represented in Table I, and the location of these measurement stations is marked in Fig. 1.

2) Satellite Observation Aerosol Optical Depth (AOD): The Moderate Resolution Imaging Spectroradiometer (MODIS) Collection 61 atmosphere products (MCD19A2) generated through the Multiangle Implementation of Atmospheric Correction (MAIAC) algorithm [25] were used as one of the most critical predictors according to previous studies [7], [8], [10], [11] to construct our ML-CNP$M_{2.5}$ sample dataset, which was

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TABLE I
DETAILED STATISTICS OF THE DATA MAKE UP ML-CNPM$_{2.5}$

Dataset	Feature	Spatial resolution (Lon × Lat)	Temporal resolution	Unit	source
Station $PM_{2.5}$	Gound-based $PM_{2.5}$	/	Hourly	$\mu g/m^3$	CNEMC
MAIAC AOD	AOD at 550nm	1 km × 1 km	Daily	/	MCD19A2
Simulation	Dust2.5 (Dust masses with size < 2.5μg)	0.5 °×0.625 °	Hourly	kg/m^3	MERRA-2
	SS2.5 (Sea-salt masses with size < 2.5μg)			kg/m^3	
	BC (Black Carbon)			kg/m^2	
	OC (Organic Carbon)			kg/m^3	
	$SO4^{2-}$ (Sulfate)			kg/m^3	
	Simulated AOD at 550nm			-	
Meteorology	BLH (Boundary layer height)	0.25°× 0.25°	Hourly	m	ERA5
	BLD (Boundary layer dissipation)			J/m	
	TEM (2m air temperature)			K	
	U10 (10m u component of wind)			m/s	
	V10 (10m v component of wind)			m/s	
	FAL (Forecast albedo)			-	
	SP (Surface pressure)			Pa	
	TCO (Total column ozone)			kg/m^2	
	TCWV (Total column water vapour)			kg/m^2	
	TPLT (Trapping layer top height)			m	
	CDIR (Clear sky direct solar radiation)			J/m	
Geography	LUC (Land use cover)	500 m×500 m	Annually	-	MCD12Q1
	NDVI (Normalized vegetation index)	1 km × 1 km	Monthly	-	MOD13A3
	DEM (Digital elevation model)	1 km × 1 km	/	m	SRTM
Economy	POP (Population data)	1 km × 1 km	Annually	count	GlobPOP
	TN (Transportation node)	/	Annually	/	OpenStreet Map

Lon: Longitude; Lat: Latitude.

based on the fact that there is a strong positive relationship between AOD and $PM_{2.5}$. The AOD data from MCD19A2 has shown excellent merits in terms of higher accuracy, more consistent with ground measurements, more results, as well as higher spatial resolution (1 km) in China compared to other aerosol products, i.e., MODIS DT, DB, and DTB products [8], thereby MCD19A2 put into our dataset. Note that only the AOD from Terra satellite with an overpass time of about 10:30 A.M. local time was collected into the first generation of ML-CNPM$_{2.5}$, and the data from Aqua satellite (overpasses ∼1:30 P.M. local time) will be added in a version update. All these AOD data were first matched with ground-based $PM_{2.5}$ stations in time and space, then added into our PD after calibrating and filling (see Section III). The details of the AOD data used, including source and spatiotemporal resolution, are listed in Table I.

3) Auxiliary Data: The chemical simulation, meteorological data, geographic data, and economic data were collected from 2014 to 2023 in China from various high-accuracy products, including the Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA-2), ECMWF Reanalysis v5 (ERA5), MCD12Q1, MOD13A3, Shuttle Radar Topography Mission (SRTM), GlobPOP, and OpenStreetMap. Since numerous studies showed that several predictors besides AOD, e.g., boundary layer height (BLH), relative humidity (RH), and surface temperature (ST), are also related to ground-level $PM_{2.5}$ concentration [24], we searched for as much comprehensive auxiliary data as possible to add to our ML-CNPM$_{2.5}$, ensuring its comprehensiveness and richness. Thus, the ML-CNPM$_{2.5}$ has the potential to train and validate multiple ML-based models that need various auxiliary data [13], [14], [17].

Specially, the chemical simulation data including dust masses, sea-salt masses, black carbon, organic carbon, and sulfate, which can be used to calculate simulated $PM_{2.5}$ concentration through (1) [26], [27], as well as AOD at 550 nm, was provided by the MERRA-2 product with a wide range of applications in $PM_{2.5}$ research [26], [27], [28], [29]. The BLH, boundary layer dissipation (BLD), 2 m air temperature (TEM), 10 m U-component of wind (10 UW), 10 m V-component of wind (10 VW), forecast albedo (FA), surface pressure (SP), total column ozone (TCO), total column water vapor

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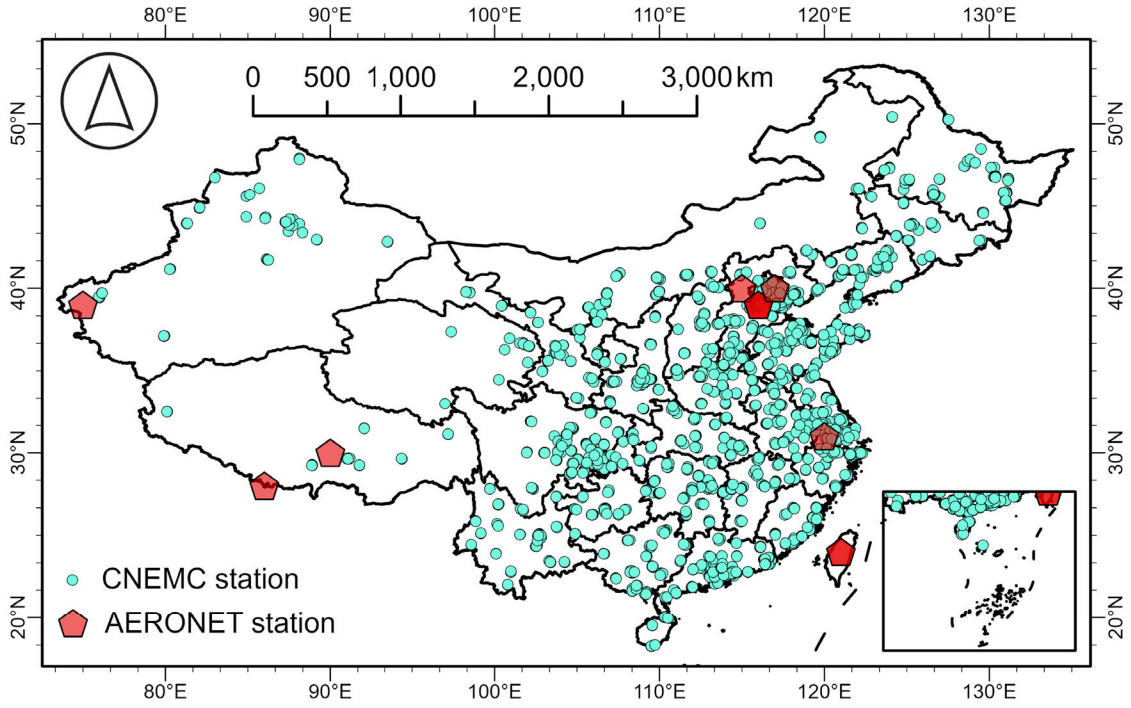


Fig. 1. Study region with ground-based $PM_{2.5}$ and AOD stations label.

(TCWV), trapping layer top height (TLTH), and clear sky direct solar radiation (SR) from ERA5, a high-precision and high-resolution atmospheric reanalysis product widely used, makes up meteorological data. Geographic data (i.e., land use cover (LUC), normalized vegetation index (NDVI), and digital elevation model (DEM) and economic data [population data (POP) and transportation node data (TN)] were selected to reflect ground conditions and human activates, respectively. In detail, annual LUC and monthly NDVI were obtained from the MODIS Level 3 product (MCD12Q1 for LUC and MCD13Q3 for NDVI). The SRTM, GlobPOP, and OpenStreetMap provided the DEM data with an uncertainty of ~0.9 m, POP data with an average relative entropy of ~0.037 from 1990 to 2020, and TND data, respectively. Table I summarizes these data information including resolution and source. Note that the POP was reprocessed by using a two-stage fitting to estimate the missing population in 2021, 2022, and 2023, as described in Text S1. The TN data was also processed in this work to generate the continuous gridded product with full coverage during 2014–2023 in China (seen in Text S2).

The above auxiliary data is all reprojected to the WGS84 coordinate system, resampled using the nearest-neighbor approach to 0.01° and clipped according to mainland China, to ensure that they can match each other, particularly with ground-based $PM_{2.5}$ measurements and satellite AOD. The specific match strategy is introduced in Text S3

$$\begin{aligned} &PM_{2.5,\text{simulated}} \\ &\quad = \text{Dust}_{2.5,\text{simulated}} + \text{SS}_{2.5,\text{simulated}} + \text{BC}_{2.5,\text{simulated}} \\ &\qquad + \text{OC}_{2.5,\text{simulated}} \times 1.6 + \text{SO}_{4\,\text{simulated}}^{2-} \times 1.375 \end{aligned} \tag{1}$$

where $PM_{2.5,\text{simulated}}$ is the simulated $PM_{2.5}$ finally treated as simulation data, and $\text{Dust}_{2.5,\text{simulated}}$, $\text{SS}_{2.5,\text{simulated}}$, $\text{BC}_{2.5,\text{simulated}}$, and $\text{SO}_{4\,\text{simulated}}^{2-}$ represent the dust masses with size greater than 2.5 μg, sea-salt masses with size greater than 2.5 μg, black carbon, organic carbon, and sulfate produced by MERRA-2, respectively.

B. Methodologies

1) AOD Calibrating According to Ground-Based AOD Measurements: The MAIAC AOD from the Terra satellite was calibrated through ground-based AOD measurements and auxiliary data using XGBoost and traditional linear regression (LR) models [7], respectively. The ground-based AOD can be provided by the AERONET sites, whose accuracy is about 0.02 for level 2, and their locations are shown in Fig. 1 with the triangles. Level 1.5 and 2 AERONET AOD in China during 2014–2023 was obtained and used to construct our calibration models (i.e., the XGBoost and LR). The uncalibrated MAIAC AOD, auxiliary data, and AERONET measurements were first matched according to a space–time matching strategy (3 × 3 pixels and one-hour duration) [30], [31]. Considering that too many features may lead to overfitting [8], [32], a procedure of feature selection was performed using the cover-importance indicator of the XGBoost model. The cover-importance of every inputted feature indicates the mean cover for the sample when the feature serves as a splitting point and can be calculated by the following equation:

$$V(k) = \frac{\sum_{t=1}^{T}\sum_{t=1}^{N(t)} I(\beta(t,i)=k) H_{\gamma(t,i)}}{\sum_{t=1}^{T}\sum_{t=1}^{N(t)} I(\beta(t,i)=k)} \tag{2}$$

where V is the importance, k is the features, T represents the numbers of total trees, $\beta(t,i)$ represents the division features

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of the ith nonleaf node of the tth tree, and $H_{\gamma(t,i)}$ represents the sum of the second-order derivatives of all samples falling on the ith nonleaf node of the tth tree.

The cover-importance score of the inputted features is demonstrated in Fig. S2, where the highest score is achieved by the longitude 814.458, whereas the lowest is latitude with a score of 83.02. This anomaly may be attributed to the study region, i.e., China, with a larger range in longitude than latitude, especially for the range of the selected AERONET stations (see Fig. 1). Therefore, the difference between ground-based AOD and longitude is more pronounced, thereby leading to a more significant impact on predicting calibrated AOD by the XGBoost model. The longitude, NDVI, day of the year (Doy), TCO, TCWV, TEM, and MAIAC AOD were selected as the inputted features to construct the XGBoost model according to their cover-importance score, whereas only the MAIAC AOD was used for the LR model. Note that the score represents only the importance of these features on ground-based AOD measurements through used AERONET stations from 2014 to 2023 in this study, without physical significance. The LR model is constructed through the slope (0.911) and offset (0.018) provided by a previous study [7], whereas the parameter of the XGBoost model was determined by using an iterative grid-search strategy. The optimal parameters obtained by the grid-search strategy are 3, 0.05, and 100 for the max depth, learning rate, and number of estimators, respectively. The remaining parameters are default in the sklearn module of Python.

To assess and compare the two models, i.e., XGBoost and LR, the uncalibrated and calibrated MAIAC AOD through XGBoost and LR models was validated against AERONET AOD, respectively. The validation results, as well as the root mean square error (RMSE), mean absolute error (AE), coefficient of determination (R^2), number (N), and the percentage of points within the expected error envelope [$\pm(0.05 + 0.2 \times$ AERONET AOD)] (WEE), are shown in Fig. 2(a). Although the MAIAC AOD has demonstrated high accuracy (RMSE = 0.134 and MAE = 0.0873) and agreement against AERONET AOD measurements (R^2 = 0.9329 and WEE = 69.81%), which is consistent with previous studies [8], it still needs to be calibrated to near AERONET AOD further because the calibrated showed higher accuracy. The calibrated result of XGBoost with RMSE of 0.9965, MAE of 0.0209, and R^2 of 0.9965 is better than that of LR with RMSE of 0.1341, MAE of 0.0851, and R^2 of 0.9329. Additionally, the results of XGBoost falling into the expected error envelope are up to 99.18%, which is significantly higher than that of RF, indicating that it is more effective using ML-based approaches to calibrate satellite AOD than using traditional regression approaches. While the LR can improve the MAIAC AOD, its calibration is restricted and not obvious, which is possibly attributed to the fact that the relationship between MAIAC AOD and AERONET AOD is not liner but is affected by other factors, such as locations, altitude, season, and vegetation cover. Therefore, the XGBoost using multiple data in addition to MAIAC AOD achieved high-accuracy calibration. Calibrated AOD using XGBoost, LR, and uncalibrated AOD are all added to our ML-CNPM$_{2.5}$ dataset. Note that the AOD in the ML-CNPM$_{2.5}$ matches the ground-based PM$_{2.5}$, not the above part of the AOD matching with AEORNET AOD measurements.

2) AOD Filling Using Chemical Simulation and Meteorological Data: In this study, the missing and invalid MAIAC AODs attributed to cloud contamination, algorithm failure, and noises were detected and filled by using chemical simulation and meteorological data. Different from filling approaches in previous studies [8], [11], we used an ML-based model (i.e., XGBoost), simulated AOD, and meteorological data to fill the Terra MAIAC AOD due to lacking Aqua satellite. The specific process is as follows.

Step 1 (Sample Construct): Data including valid MAIAC AOD, simulated AOD, TEM, TCO, TCWV, DEM, NDVI, LON, LAT, and Doy in China during 2014–2023 was obtained to make up our sample according to the time, longitude, and latitude of the simulated AOD.

Step 2 (Feature Selection): Considering the cover-importance indicator, feature sizes, and previous studies [7], [8], [33], nine features, including simulated AOD, TEM, TCO, TCWV, DEM, NDVI, LON, and Doy, were selected as the input of the XGBoost model. The detailed score of these features is shown in Fig. S2. Since the selected features with relatively high cover-importance scores had been widely used in AOD retrieval and filling studies [8], [33], [34], AOD gaps in the MAIAC product can be effectively filled with high accuracy.

Step 3 (Model Construction): The grid-search strategy is also used, to obtain the optimal parameters for XGBoost in terms of the max depth (3), learning rate (0.01), and number of estimators (300). The XGBoost model is constructed using the sklearn module in Python. For comparison, the LR model is also fitted using (3).

Step 4 (AOD Filling): The fitted LR and trained XGBoost were used to fill the missing MAIAC AOD from 2014 to 2023 in mainland China. Note that the gap-free AODs keep their original values, yet missing areas are filled with new values generated by the LR and XGBoost models.

Step 5 (Model Validation): The filled AOD was validated by the AERONET AOD to assess the performance of our filling methods. The high-accuracy AOD can ensure the accuracy of the following PM$_{2.5}$ retrieval:

$$\mathrm{AOD}^{L}_{\mathrm{filled}} = k \times \mathrm{AOD}_{\mathrm{MERRA}} + b \tag{3}$$

where $\mathrm{AOD}^{L}_{\mathrm{filled}}$ and $\mathrm{AOD}_{\mathrm{MERRA}}$ represent the AOD filled by using the LR model and the AOD from MERRA simulation, respectively. k and b are the fitting factors, with values of 1.031 and 0.028, respectively. The two fitting factors were obtained by fitting the overall samples using a least square method in the numpy module of Python.

Fig. 2(d)–(f) shows the validated scatter points, RMSE, MAE, R^2, N, and WEE. The filled by XGBoost have similar accuracy in terms of MAE (0.098), R^2 (0.87), and WEE (82.69%) compared to the unfilled [see Fig. 2(c)], whose MAE, R^2, and WEE are 0.0209%, 0.9965%, and 99.18%, respectively, while more discrete results of XGBoost. More importantly, all indicators are improved significantly (~16% for R^2, ~24.6% for RMSE, ~46.4% for MAE, and ~55.04 for

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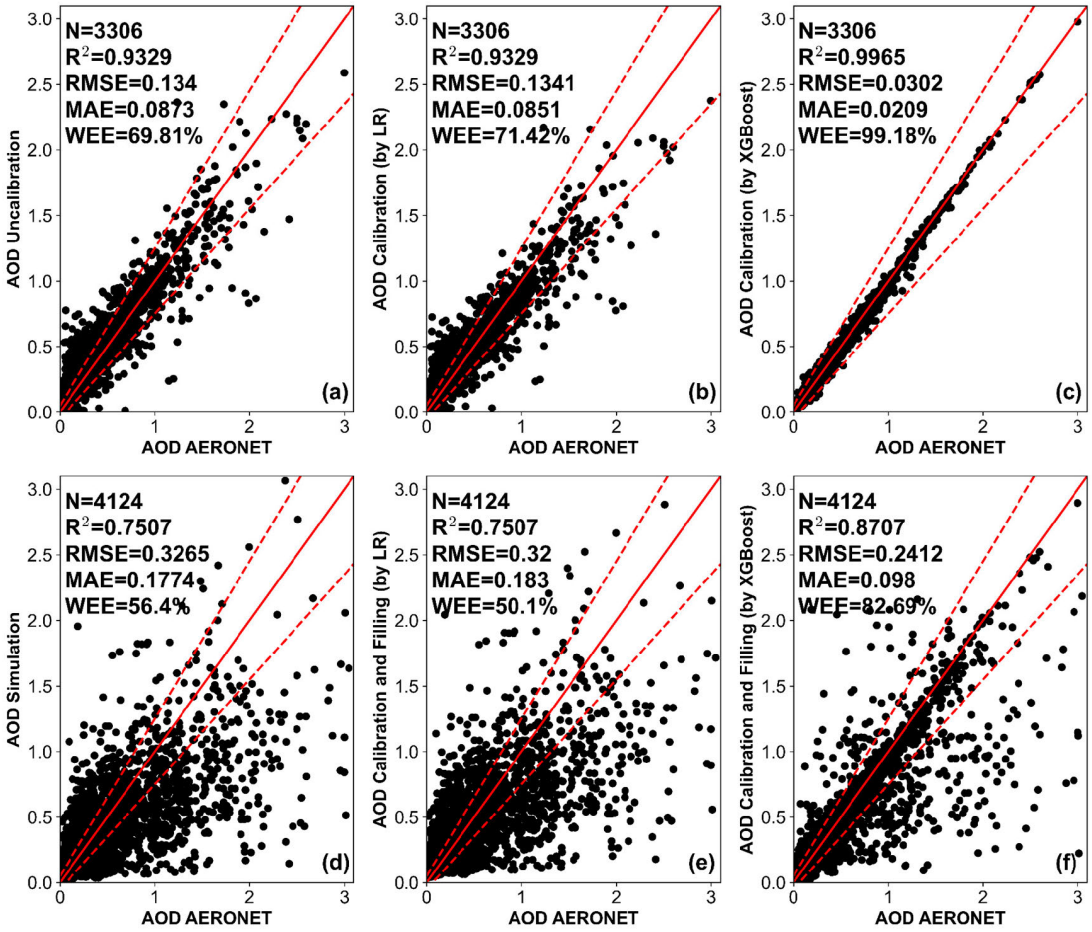


Fig. 2. Validation of (a)–(c) AOD calibration and (d)–(f) AOD filling. (a) MCD19A2 Terra AOD and (d) MERRA-2 AOD. (b) and (e) LR model calibration and filling. (c) and (f) XGBoost model calibration and filling.

WEE) by the XGBoost compared to that of LR with RMSE of 0.2412, MAE of 0.183, R^2 of 0.7507, and WEE of 50.1%. The advantage of ML based on filling missing AOD is also proven there. Notably, thanks to our filling method, more number (4124) was found in the filled AOD than that of the unfilled (3306) in this sample, significantly increasing samples in our validating dataset. Actually, the above filling only occurred in those scenes without clouds because the sun photometer in AERONET stations only operates under clear skies, thereby being validated there. However, our filled model can never be applied to any conditions, regardless of how many missing AODs. Take an example: our modes (XGBoost and LR) were applied to the MAIAC AOD under different periods, i.e., January 4, 2014, October 30, 2019, and June 15, 2023, and their results are shown in Fig. 3. Under the various valid values from 30.29% to 58.94%, both LR and XGBoost filled the missing AOD successfully, especially for XGBoost with over 100% filling. Additionally, the filled results from these models reconstructed the AOD spatial distribution well, aligning with the trend of the unfilled AOD. However, the LR demonstrated a significant mosaic phenomenon, which may be attributed to the fact that the LR only uses the simulated AOD with coarse resolution (0.5°–0.625°) to fill the fine MAIAC AOD (0.01°), thereby leading to significant coarse pixels and filling cover of less than 100%, particularly in locations where the AOD is lacking. In contrast, the XGBoost fully utilized multiple factors, including simulated AOD and meteorological and geographical data, thereby causing better filling results in spatial patterns, benefitting from the solid data-fitting ability of the ML-based model. It is noted that the filled and unfilled AODs matching the ground-based $PM_{2.5}$ measurements were all included in ML-CNP$M_{2.5}$, supporting various research uses for different purposes.

3) Basic Models: Eight basic models, including RF, RT, ERT, XGBoost, LightGBM, DBN, FCNN, and ResNet, were widely used in estimating $PM_{2.5}$. We selected them to be trained and validated through our ML-CNP$M_{2.5}$ to demonstrate the viability and comprehensiveness of our proposed sample dataset.

These ML-based models can be classified into two categories: tree-based models (i.e., RF, ERT, XGBoost, LGB, and AdaBoost) and neuron-based models (DBN, FCNN, and ResNet). The tree-based models use decision trees as the basic modules to learn and fit samples, whereas the neuron-based models are based on the neurons. These various ML-based models can effectively assess our ML-CNP$M_{2.5}$ due to their distinct fundamental learning strategies. Since our object focuses on the ML-CNP$M_{2.5}$ generation rather than

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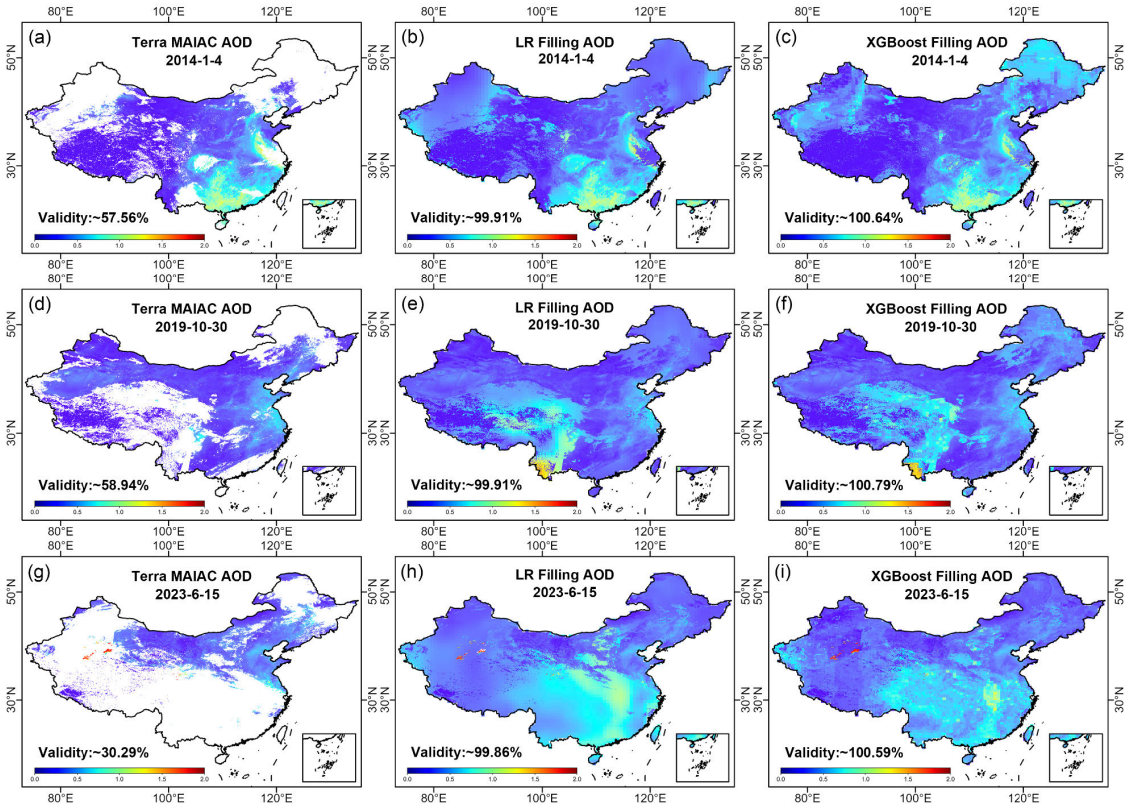


Fig. 3. Fill-AOD spatial pattern of LR and XGBoost under different periods. (a)–(c) January 4, 2014. (d)–(f) October 30, 2019. (g)–(i) June 15, 2023. Validity represents the proportion of valid AOD pixels to all pixels.

the optimal ML-based model building, only the most basic ML models were constructed in this study. More importantly, we want to train these models so basic that they can serve as the baseline to judge other derived or new models in subsequent studies. These basic validated models will also be released, accompanied by the ML-CNPM$_{2.5}$. Considering the time consumed and these basic models, fundamental constructs and parameters were used to build these models. In detail, the tree-based models (i.e., XGBoost, LGB, RF, ERT, and AdaBoost) were designed based on the default parameters in the sklearn module of Python, whereas the neuron-based models used the simple construct shown in Table S2.

4) Model Validation: The ten-fold cross-validation (10 CV) approach was used to validate these basic ML-based models, i.e., RF, RT, ERT, XGBoost, LightGBM, DBN, FCNN, and ResNet. The 10 CV has been widely used in the field of PM$_{2.5}$ retrieval using satellite remote sensing to assess the ability of fitting models, especially ML-based models [8], [12], [23], [24], [35]. For the process of 10 CV, samples are first randomly divided equally into ten groups; then, one group is selected randomly to validate the model after training through the remaining nine groups and repeat training and validation until every group has been used to validate the model. Finally, the average of the above ten validation results is the indicator assessing the model.

In this study, to simplify, the calibrated momentary average (9:00–12:00 A.M. local time) in our ML-based CNPM$_{2.5}$ rather than the filled with larger sample size was used to train these ML-based models and validate them using 10 CV. The special results, including quantitative scatterplot and error distribution, are demonstrated in Section III.

III. Results

A. *Ml-CNPM$_{2.5}$*

The features possibly affecting ground-based PM$_{2.5}$ from 2014 to 2023 in China were collected to make up our first version of the ML-CNPM$_{2.5}$. Thanks to our filling and calibrating methods, over five million samples (5 076 608) have been obtained, which is so more PM$_{2.5}$ samples that have not been covered in previous studies, to our knowledge. To train and assess different models in terms of primary and higher accuracy ML-based models, the dataset including unfilled AOD, with 1 790 210 line records, is also issued since filled AOD always shows lower accuracy than unfilled. To distinguish the two datasets, the filled AOD dataset is named ML-CNPM$_{2.5}$-A and the unfilled is named ML-CNPM$_{2.5}$-B. There are 24 features contained in the ML-CNPM$_{2.5}$-A, whereas 23 features in the ML-CNPM$_{2.5}$-B. Most of the features directly affect or indirectly affect ground-based PM$_{2.5}$ estimating using remote sensing and ML technology, thereby being widely used as the input of ML-based models [8], [12], [14], [23]. The distribution of each feature in the ML-CNPM$_{2.5}$-A (CNPM$_{2.5}$-B) is revealed in Fig. 4 (see Fig. S3). The figures intuitively demonstrate each feature's range of values, including median, quartile, and outlier. For example, the distribution of Terra MAIAC AOD is changed plainly after being calibrated, i.e., from the range of 0–8 calibrated to the range of 0–3, which is more realistic. The

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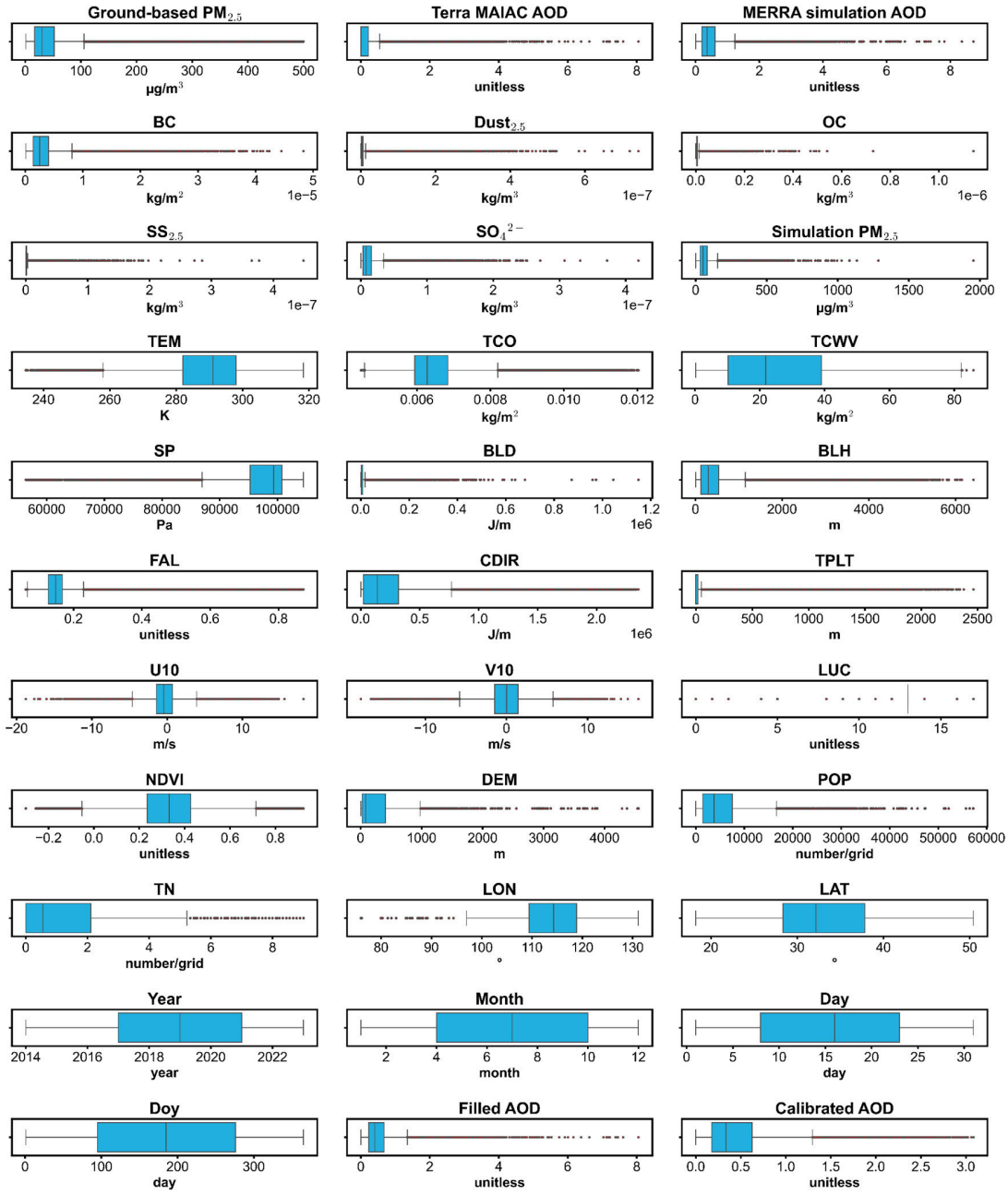


Fig. 4. Distribution of features made up ML-$CNPM_{2.5}$ with AOD calibrating and filling.

discrete features, including year, month, day, Doy, and LUC, show even distribution in their range of values, indicating the equilibrium and comprehensiveness of our sample dataset. Detailed information about these features is listed in Table II (see Table S1) for ML-$CNPM_{2.5}$-A ($CNPM_{2.5}$-B). Overall, our sample dataset includes commend features used widely in estimating $PM_{2.5}$, with high volume and comprehensive records, as big data ensures the training and validation of different models.

B. Validation and Comparison of Basic ML-Based Models

1) Quantitative 10 CV Results: Taking advantage of our ML-$CNPM_{2.5}$, the eight basic ML-based models have been trained and validated using 10 CV. Fig. 5 shows the performance of these models on estimating ground-based $PM_{2.5}$, including density scatterplots and assessment indicators (i.e., MAE, RMSE, R^2, and number). Surprisingly, although these models were constructed using default or simple parameters, most of them, except for the AdaBoost model, have been trained well and showed high accuracy and estimating ability in 10 CV, possibly attributed to our ML-$CNPM_{2.5}$ dataset. The ERT model demonstrates the best performance proven by the accuracy with MAE of 11.2115 μg/m^3 and RMSE of 18.1799 μg/m^3, and by the estimating power with R^2 of 0.8868, due to its excellent default structure with powerful fitting ability, which is consistent with the findings in previous

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TABLE II
DETAIL STATISTICS OF THE FEATURES IN ML-CNPM$_{2.5}$ WITH AOD CALIBRATING AND FILLING

Feature	Max	Min	Mean	Median	Unit	Total Number
Ground-based $PM_{2.5}$	499.75	1	41.3372	30.25	$\mu g/m^3$	
Terra MAIAC AOD	8.0497	0	0.1431	0	unitless	
MERRA simulation AOD	8.723	0.0061	0.4518	0.3596	unitless	
BC	4.83E-05	8.59E-08	3.05E-06	2.52E-06	kg/m^2	
Dust2.5	7.43E-07	8.95E-19	5.61E-09	2.71E-09	kg/m^3	
OC	1.14E-06	1.94E-12	5.78E-09	4.79E-09	kg/m^3	
SS2.5	4.49E-07	8.30E-20	1.26E-09	3.39E-10	kg/m^3	
SO4	4.19E-07	8.45E-13	1.10E-08	8.33E-09	kg/m^3	
Simulation $PM_{2.5}$	1954.719	1.1044	61.7921	52.8109	$\mu g/m^3$	
TEM	318.2958	234.6529	289.2147	291.0471	K	
TCO	0.012032	0.004471	0.006479	0.006282	kg/m^2	
TCWV	85.9099	0.2471	25.8178	21.8325	kg/m^2	
SP	104511.7	56364.08	96312.72	99358.39	Pa	
BLD	1148877	20.252	7245.596	3983.758	J/m	
BLH	6414.364	9.1091	421.6479	299.8185	m	5,076,608
FAL	0.8727	0.06	0.1561	0.1475	unitless	
CDIR	2354575	0	233385.8	139713.7	J/m	
TPLT	2462.449	-1	30.2382	-1	m	
U10	18.1116	-18.779	-0.3144	-0.4337	m/s	
V10	16.2722	-17.9446	-0.0448	0.026	m/s	
LUC	17	0	12	13	unitless	
NDVI	0.9227	-0.3	0.3381	0.3313	unitless	
DEM	4553.334	4.5	377.7322	86.5556	m	
POP	57268	0	5529	3758	number/grid	
TN	9	0	1.4564	0.5556	number/grid	
LON	131.1652	75.9538	113.4599	114.3014	°	
LAT	50.4278	18.2489	33.0642	32.1875	°	
Year	2023	2014	2019	2019	year	
Month	12	1	7	7	month	
Day	31	1	16	16	day	
Doy	365	1	185	185	day	
Filled AOD	8.0497	0.0001	0.5076	0.4064	unitless	
Calibrated AOD	3.084	0.0001	0.4738	0.3344	unitless	

studies [7], [8]. The following model is the RF model, which has a construct similar to that of the ERT, with similar MAE (11.691 $\mu g/m^3$), RMSE (18.9345 $\mu g/m^3$), and R^2 (0.8766). The worst performing model is the AdaBoost, showing large uncertainty in terms of MAE of 35.7935 $\mu g/m^3$, RMSE of 47.2989 $\mu g/m^3$, and R^2 of 0.5555, which may be due to its simple construct and first proposed for classification. The neuron-based models, however, demonstrate lower accuracy (MAE 16–17 $\mu g/m^3$, RMSE 27–28 $\mu g/m^3$, and R^2 0.70–0.73) than the tree-based models except for the AdaBoost model, similar to some studies [16], [17], which may be attributed to their simple constructs used in this study. In fact, by careful setup and adjustment of parameters, the neuron-based models would show better performance, as proved by the different results of FCNN-256 (with 256 neurons in the first layer) and FCNN-512 (with 512 neurons in the first layer), even only changing 256 neurons. The 10 CV results of our basic models and relevant studies are shown in Table III, where our ERT demonstrates similar estimating uncertainty with the optimal model (i.e., STET) with $R^2 = \sim 0.9$ and MAE $<$ 0.12 $\mu g/m^3$. However, compared to previous studies, our ERT has been trained and validated using more samples (1 790 210) during the most extended period (2014–2023) based on our ML-CNPM$_{2.5}$. In general, eight basic ML-based models as the baseline of $PM_{2.5}$ estimating have been given in this study by using our ML-CNPM$_{2.5}$ with a maximum number of

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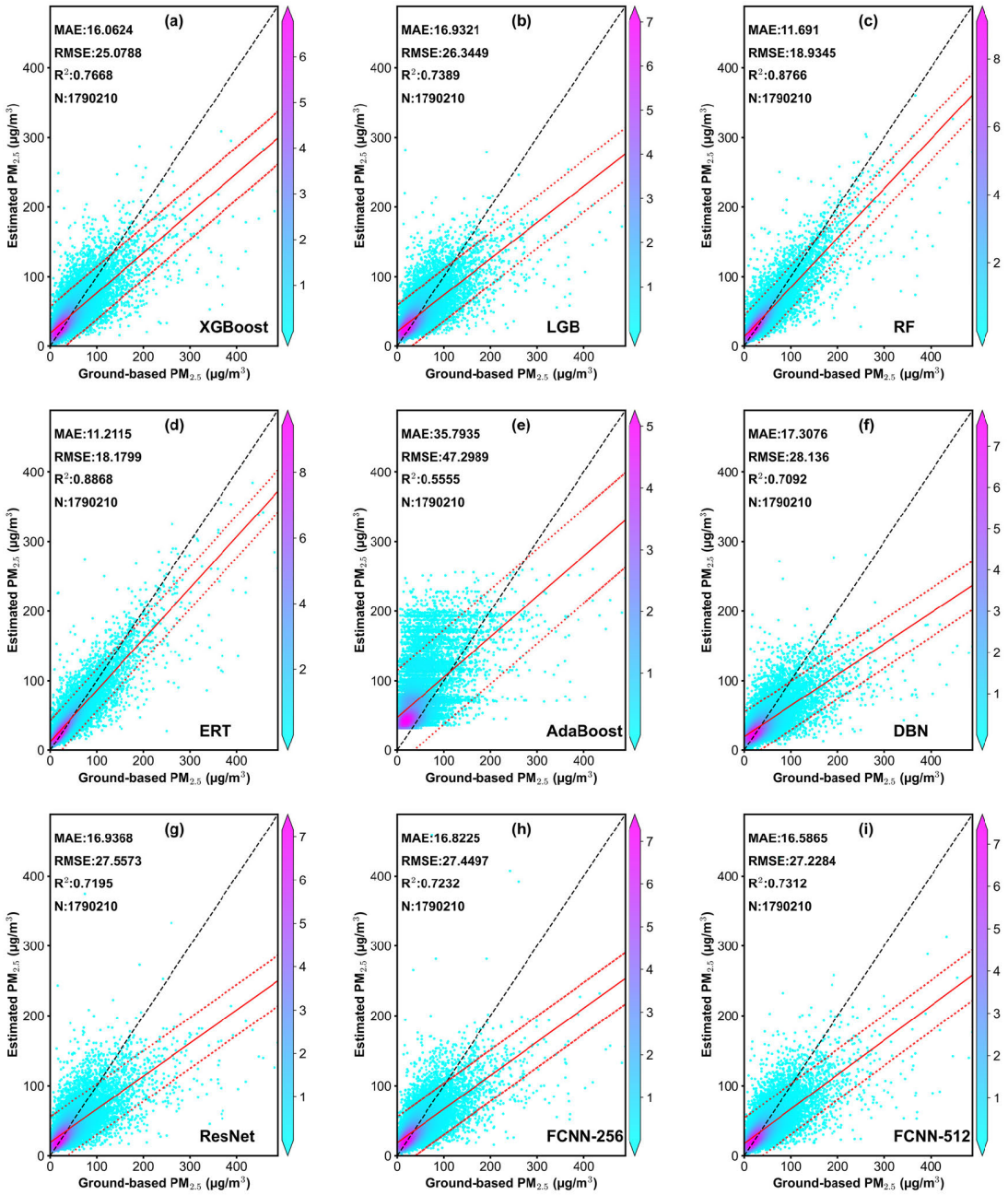


Fig. 5. Density scatterplot of $PM_{2.5}$ estimated by different models [(a) XGBoost, (b) LGB, (c) RF, (d) ERT, (e) AdaBoost, (f) DBN, (g) ResNet, (h) FCNN-256, and (i) FCN-512] against ground-based $PM_{2.5}$. The black line, red line, and red dotted line represent the 1:1 line, linearly fitting line, and 95% CI.

data records and periods. The excellent performance of these models suggests that our sample dataset can be used to train and validate ML-based models effectively.

2) Spatial Uncertainty of Basic Models: The uncertainty regarding RMSE and R^2 of different models obtained by the 10 CV from 2014 to 2023 is shown in Fig. 6 according to peer stations with at least ten measurements for statistical significance. From Fig. 6, small estimating biases with RMSE < 25 μg/m^3 can be seen in most stations (93%–59%) for different models except for the AdaBoost model with most sites ($\sim$91%) of RMSE > 30 μg/m^3. These models all estimate $PM_{2.5}$ most with more significant error in central and eastern China, where frequent haze and dust storms are common due to complex particulate emissions, transport, and deposition during the study period, especially before 2008 [8], [36]. Not only our basic models but also other designed models show similar RMSE distribution in China [7], [8], [13].

Therefore, further efforts are needed to find the reason why these ML-based models show low accuracy there, improving their flexibility in space. In contrast, for R^2, which indicates the estimating power of ML-based models, large values of >0.75 are observed in central and eastern China for each model, whereas the smaller ones are in north-western and southern China. This anomaly has also been seen in previous studies [8], possibly due to the uncertainty of Terra MAIAC AOD and other meteorological conditions (e.g., BLH) in

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TABLE III
PERFORMANCE OF DIFFERENT PM$_{2.5}$-ESTIMATING MODELS USING REMOTE SENSING TECHNOLOGY INCLUDING THIS STUDY AND PREVIOUS STUDIES

Model	MAE ($\mu g/m^3$)	RMSE ($\mu g/m^3$)	R^2	N	Date	Reference
GWR	21.25	32.98	0.64	58,164	2012-2013	[6]
GW-GBM	23	-	0.76	103,071	2014	[35]
Geoi-DBN	13.03	8.54	0.88	-	2015	[13]
RF	28.1	-	0.83	96,103	2014–2016	[23]
CMAQ	17.7	27.8	0.606	407,049	2013-2015	[12]
RF	11.50	17.91	0.81	153,648	2015-2016	[7]
STRF	9.77	15.57	0.85			
STET	9.147	14.6	0.89	834,737	2013-2018	[8]
SIDLM	15.69	-	0.68	118,133	2016-2019	[14]
XGBoost	16.06	25.08	0.77			
LGB	16.93	26.34	0.74			
RF	11.69	18.93	0.88			
ERT	11.21	18.18	0.89			
AdaBoost	35.79	47.30	0.56	1,790,210	2014-2023	This study
DBN	17.31	28.14	0.71			
ResNet	16.94	27.56	0.72			
FCNN-256	16.82	27.45	0.72			
FCNN-512	16.59	27.22	0.73			

GWR: Geographically Weighted Regression; GW-GBM: Geographically-Weighted Gradient Boosting Machine; Geoi-DBN: Geographical distance and spatiotemporal deep belief network; RF: Random Forest model; STRF: Space-Time Random Forest model; STET: Space-Time Extra-Trees model; SIDLM: Spatial-Temporal Interpretable Deep Learning Model.

north-western and southern China, which also need to be addressed in future work. In a word, the ML-based models trained by using our ML-CNPM$_{2.5}$ show similar uncertainty in ground-based stations in China with previous studies, which suggests that our sample dataset is meaningful for training different models and there are same issues like other studies should be addressed in the coming work.

3) Spatial Pattern of Estimated PM$_{2.5}$ Under Severe Haze: A practical model can not only precisely estimate PM$_{2.5}$ in points but can also appropriately capture spatial trends in PM$_{2.5}$. Thus, these basic models estimated PM$_{2.5}$ on December 28, 2023, when most areas of China suffered from severe haze, to assess their estimating ability in space. Their results are shown in Fig. 7, except for the results of the AdaBoost model due to its poor performance (see Fig. S4). A similar spatial pattern of PM$_{2.5}$ is observed in most models, with large values of PM$_{2.5}$ > 120 $\mu g/m^3$ in central and south-western China, especially in Shandong, Henan, Hebei, Chongqing, and Sichuan, which is consistent with the actual situation at the time. The tree-based models tend to estimate higher PM$_{2.5}$ than the neuron-based models, thereby causing shallower color in Fig. 7. Additionally, the mosaic effect is seen in XGBoost [see Fig. 7(a)] and AdaBoost (see Fig. S4), attributed to the too much reliance on the location information of grids to obtain accurate spatial distribution information of PM$_{2.5}$. In contrast, other models show fine spatial patterns, especially the ERT [see Fig. 7(d)], DBN see [Fig. 7(f)], and FCNN-512 [see Fig. 7(h)] models. More importantly, different from other models [8], [14], our models can obtain daily fill-cover PM$_{2.5}$ by using our filling method and ML-CNPM$_{2.5}$ dataset, no need to average different sources of AOD (e.g., Terra and Aqua AOD) [11]. Overall, according to our ML-CNPM$_{2.5}$, our trained models all successfully capture spatial trends in PM$_{2.5}$ and have estimated PM$_{2.5}$ distribution in space consistent with the actual conditions. However, some models, XGBoost and AdaBoost, show apparent mosaic effects that may be mitigated using reasonable parameters and structure.

IV. DISCUSSION

A. What Can Our ML-CNPM$_{2.5}$ Do?

The ML-CNPM$_{2.5}$, as a sample dataset for PM$_{2.5}$ estimating using remote sensing and ML-based technology, can be used to train different models effectively due to its large column size (over five million) and various features from different factors related to chemical simulation, meteorology, geography, and economics. The above-trained basic models exemplify how our ML-CNPM$_{2.5}$ can be applied for model training. In addition, the ML-CNPM$_{2.5}$ can help validate different PM$_{2.5}$ estimating models objectively and comprehensively since it covers even and rich situations, i.e., various PM$_{2.5}$ concentrations and features. Based on our dataset, different novel models proposed in subsequent studies by others can be validated under a unified standard, ensuring the comparability of different models. Our dataset can also perform feature selection and uncertainty analysis for different features inputted into

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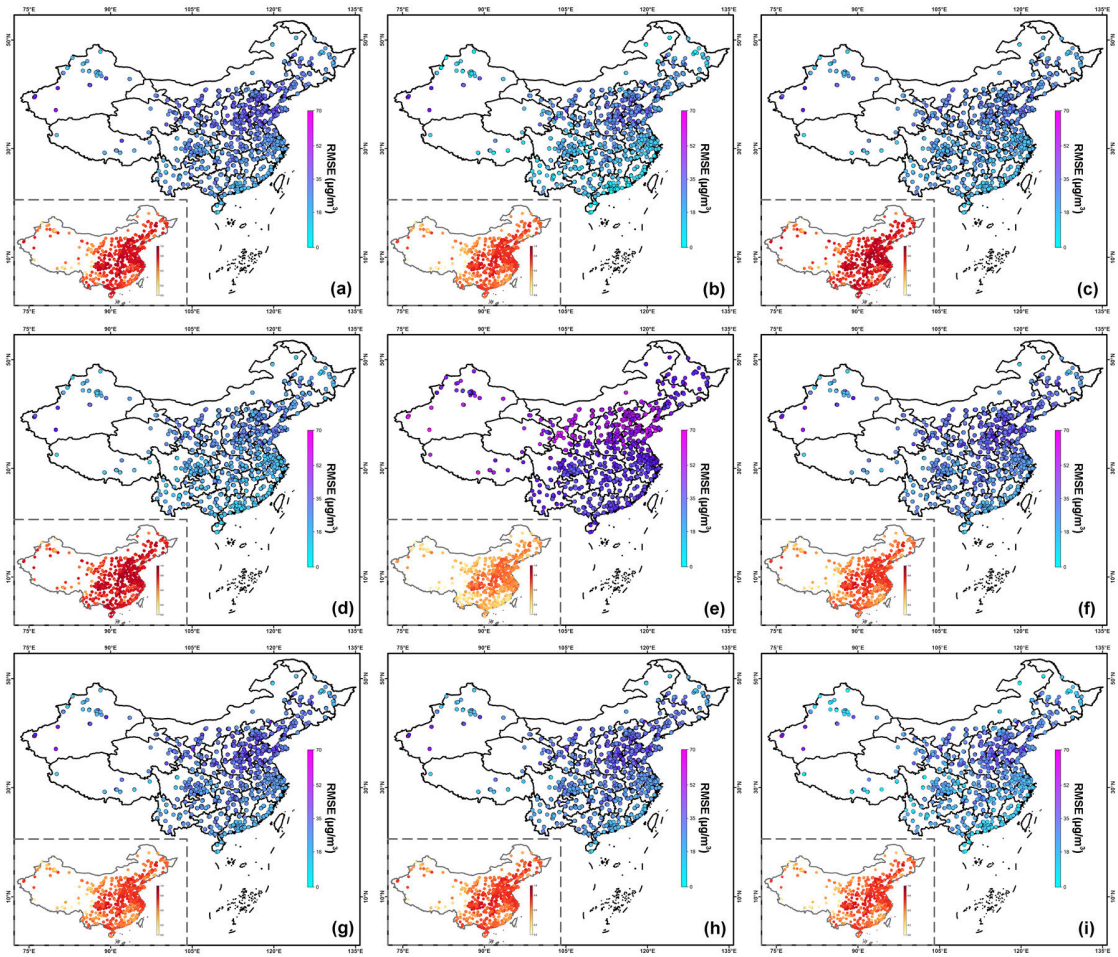


Fig. 6. Spatial uncertainty of different models (i.e., (a) XGBoost, (b) LGB, (c) RF, (d) ERF, (e) AdaBoost, (f) DBN, (g) ResNet, (h) FCNN-256, and (i) FCNN-512) using ten CV from 2014 to 2023. The large graph represents the RMSE distribution, and the small graph in the lower left corner represents the R^2.

models [8]. For instance, our ML-CNPM$_{2.5}$-B is applied to determine the importance of different features in estimating PM$_{2.5}$ using the cover-importance [calculated by (2)] and gain-importance [calculated by (S1)] indicators. The results are shown in Fig. S5. Unexpectedly, their importance is changed as different indicators, but only one indicator, were used to select the inputted features into their ML-based models in most studies [7], [8], [11], including this study. Therefore, a significant uncertainty of feature selection possibly existed when using them to construct models. Additionally, these importance indicators only serve as the basis for selecting appropriate features for a particular model, e.g., XGBoost. Thus, a more robust approach to select features should be developed to balance the model's accuracy and interpretability. Here, our ML-CNPM$_{2.5}$ can serve as the fundamental feature dataset to assess these approaches designed for feature selection.

B. What Problem Our Dataset Solves?

ML-based models have become one of the most important ways to obtain high accuracy and large-scale PM$_{2.5}$ maps. However, the quantity, quality, and source of the samples directly affect the performance of the ML-based models since these models require supervised training in advance. Previous studies had privately constructed different samples with various features and trained different models based on these samples. Therefore, their models and results are incomparable due to inconsistent samples. Since our ML-CNPM$_{2.5}$ was publicly constructed using multisource data, including satellite observation, chemical simulation, meteorological data, geographical data, and economic data, it can be used to train comparable ML-based models and validate their performance in PM$_{2.5}$ retrieval. Our dataset can help researchers directly train their developed models and avoid the expense of constructing samples. Besides, the trained models by our dataset can serve as a baseline for evaluating models in other studies.

C. Why Can These Models Serve as a Baseline?

In this study, two categories of ML-based models, i.e., tree-based and neuron-based models, including XGBoost, LGB, RF, ERT, AdaBoost, DBN, ResNet, and FCNN, have been constructed and trained by our publicly available ML-CNPM$_{2.5}$. Therefore, these models are comparable and public, meaning they can estimate PM$_{2.5}$ in China by inputting suitable features. In addition, due to their relatively high precision and estimation capability, especially the ERT model, they can serve as the basic models to assess the novel models proposed

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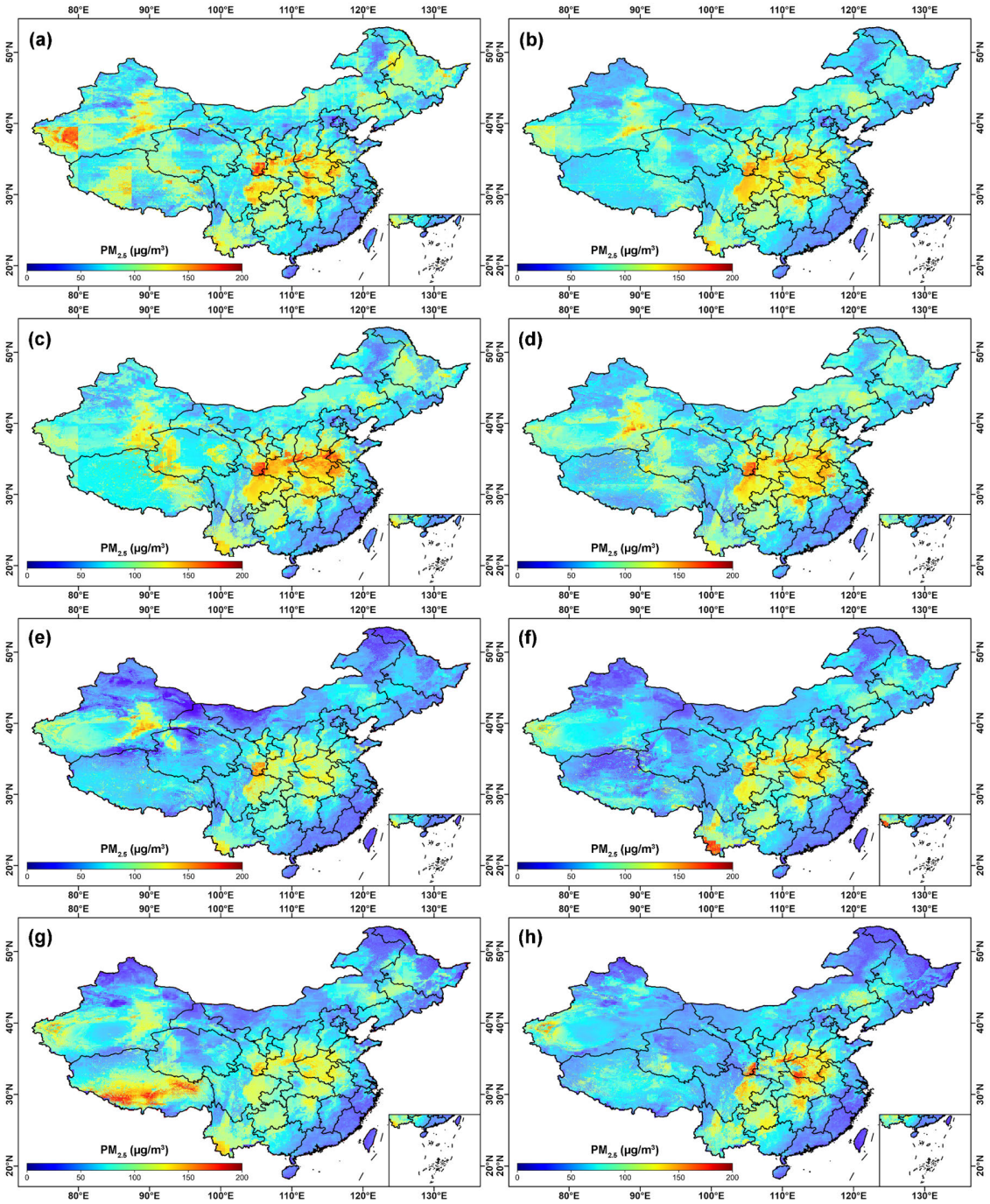


Fig. 7. Spatial $PM_{2.5}$ estimated by different models (i.e., (a) XGBoost, (b) LGB, (c) RF, (d) ERF, (e) DBN, (f) ResNet, (g) FCNN-256, and (h) FCNN-512) on December 28, 2023, when suffering with severe haze nationally.

in different studies objectively and reasonably, especially to validate their advantages. Not only their accuracy but also their time consumption when training is also given as references. The time consumption of different models at each 10 CV is shown in Fig. S6. Furthermore, these models were all trained using simple or default parameters, resulting in a small number of parameters and time consumption that different scholars can easily reproduce. These basic models trained by our ML-CNPM$_{2.5}$ can be used as the baseline to assess other models designed for $PM_{2.5}$ estimating based on remote sensing technology.

D. What Are the Issues That Need to Be Addressed Urgently?

While constructing the ML-CNPM$_{2.5}$ and training basic models, we found multiple challenges that need to be overcome to further improve the accuracy and credibility of spatial monitoring of $PM_{2.5}$. First, our sample dataset only includes the sample in mainland China, meaning that the models are trained using our ML-CNPM$_{2.5}$ only to estimate $PM_{2.5}$ of mainland China. Therefore, whether they are applied to oceans and other countries is uncertain. Likewise, our sample dataset will probably only be used to validate those models constructed in China. However, this issue can be

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addressed by adding samples under other regions, supporting the construction of different models. These new samples will be added in the new release as planned! In addition, other AOD sources, e.g., Aqua AOD, VIIRS AOD, GK2A AOD, and Himawari-8 AOD, as well as AOD at different wavelengths, e.g., 470 nm, would be added to our ML-CNPM$_{2.5}$ for further improving accuracy and contiguity of PM$_{2.5}$ estimating. Last but not the least, based on our sample dataset, the features inputted into models should be carefully considered using different analytical methods, improving the robustness and interpretability of ML-based models.

V. Conclusion

A large and feature-rich sample dataset, namely ML-CNPM$_{2.5}$, has been constructed in this study. The dataset includes ground-based PM$_{2.5}$ measurements and other features, including satellite observation, chemical modeling, and meteorological, geographic, and economic data from 2014 to 2023 in mainland China. Multiple methods were applied to construct the sample dataset. Mainly there, an ML-based method was proposed to calibrate and fill missing AOD in satellite observation due to clouds and high-light surfaces. A two-stage fitting method was used to estimate the missing population in 2021, 2022, and 2023, and a novel method was used to generate continuous gridded transportation node data. The detailed process of the CNPM$_{2.5}$ construction is fully shown in this article, and the constructed CNPM$_{2.5}$ dataset has been made public on https://doi.org/10.57760/sciencedb.08635. To our knowledge, the dataset is the most data-recorded, characterized, and time-spanning sample data available. The ML-CNPM$_{2.5}$ can be used objectively and adequately to train, validate, and compare different ML-based models designed for PM$_{2.5}$ retrieval in China based on remote sensing technology. Besides, by using the dataset, eight basic models (i.e., XGBoost, LGB, RF, ERT, AdaBoost, DBN, ResNet, and FCNN) have been trained to serve as the baseline for assessing other new models for PM$_{2.5}$ estimating. Validation results suggested that the ERT model with default parameters performed better in quantitative indicators and space patterns than other models. We also found that the deep learning models, i.e., DBN, ResNet, and FCNN, did not show better results using this sample dataset compared to the traditional ML-based models. These models have been made public with the dataset.

Overall, our public ML-CNPM$_{2.5}$ sample dataset has essential implications for developing and improving models for PM$_{2.5}$ estimating using remote sensing technology, helping monitor PM$_{2.5}$ more accurately and continuously from space.

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